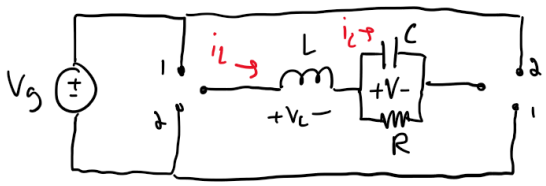
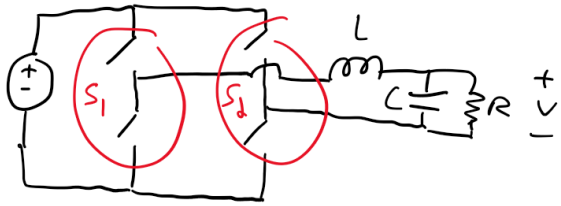
Erickson 2.5 : H-Bridge converter - conversion ratio

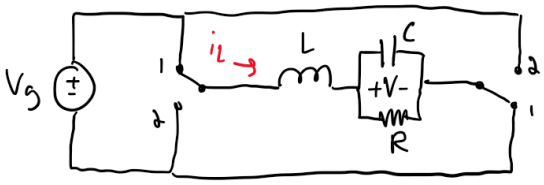
Task: find $M(D) = \frac{V}{V_g}$ and sketch $M(D)$ vs. D

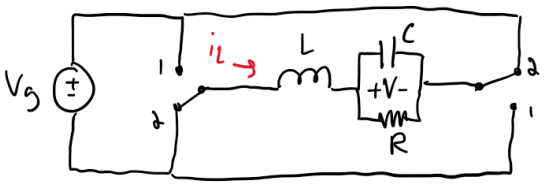


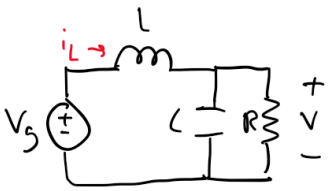
State 1: $0 < t < DT_s$ Position 1



State 2: $DT_s < t < T_s$ Position 2

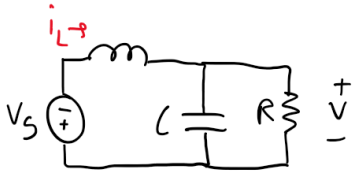






$V_L = V_g - V$

$i_C = i_L - \frac{V}{R}$



$V_L = -V_g - V$

$i_C = i_L - \frac{V}{R}$

small-ripple approximation: $v(t) = V$: output voltage ripple

$$0 = \frac{1}{T_s}\left[\int_0^{DT_s} (V_g - V)\,dt + \int_{DT_s}^{T_s} (-V_g - V)\,dt\right]$$

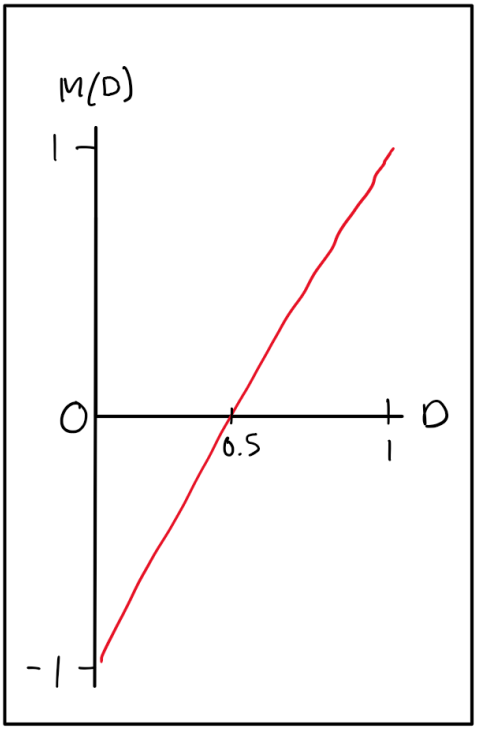
$$0 = \frac{1}{T_s}\cdot\left(DT_s(V_g - V) + T_s(-V_g - V) - DT_s(-V_g - V)\right)$$

$$0 = DV_g - DV - V_g - V + DV_g + DV$$

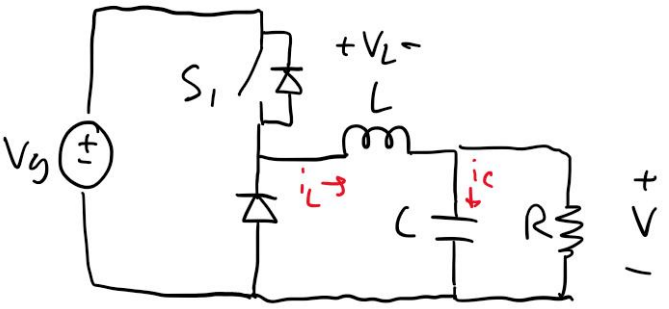
$$0 = 2DV_g - V_g - V$$

$$V = 2DV_g - V_g$$

$$\boxed{M(D) = \frac{V}{V_g} = 2D - 1}$$



Erickson 5.19 : Buck converter boundry mode on the V_g - P_{load} plane

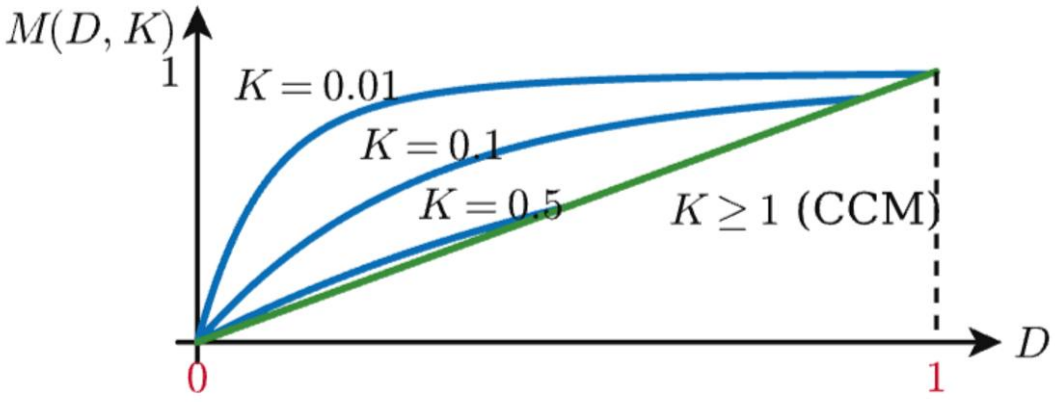


$35V \leq V_g \leq 70V$

$V = 28V$ using closed-loop control

$10W \leq P_{load} \leq 1000W$

$L = 22\mu H$, $C = 470\mu F$, $f_{sw} = 75kHz$



Task : a) determine CCM ranges for V_g & P_{load}

b) determine min/max D values @ steady state

$$P_{load} = \frac{V^2}{R} = \frac{28^2}{R} = \frac{784}{R}$$

From slides: $K = \frac{2L}{RT_s}$ $\quad M_{CCM} = D$

$$M_{DCM} = \frac{2}{1+\sqrt{1+\frac{4K}{D^2}}}$$

Boundary is when $M_{DCM} = M_{CCM}$

$$\frac{2}{1+\sqrt{1+\frac{4K}{D^2}}} = D, \quad D_{ccm} = \frac{V}{V_g}, \quad R = \frac{V^2}{P}$$

$$\frac{V}{V_g} = \frac{2}{1+\sqrt{1+\frac{4K}{\left(\frac{V^2}{V_s^2}\right)}}}$$

$$V + V\sqrt{1+\frac{4K}{V^4/V_s^2}} = 2V_g$$

$$V + V\sqrt{\frac{V^2+4KV_s^2}{V^2}} = 2V_g$$

$$V + \sqrt{V^2+4KV_s^2} = 2V_g$$

$$\sqrt{V^2+4KV_s^2} = 2V_g - V$$

$$V^2+4KV_s^2 = (2V_g-V)^2$$

$$V^2+4KV_s^2 = 4V_g^2+V^2-4VV_g$$

$$\cancel{4}K\cancel{V_s^2} = \cancel{4}V_s^2 - \cancel{4}VV_s$$

$$KV_s^2 = V_s^2 - VV_s$$

$$\frac{KV_s}{V_s} = \frac{V_s}{V_s} - \frac{V}{V_s}$$

$$K = 1 - \frac{V}{V_s}$$

$$\frac{2L}{RT_s} = 1 - \frac{V}{V_s}$$

$$P = \frac{V^2}{R} \quad \therefore R = \frac{V^2}{P}$$

$$\frac{2LP}{V^2 T_s} = 1 - \frac{V}{V_s}$$

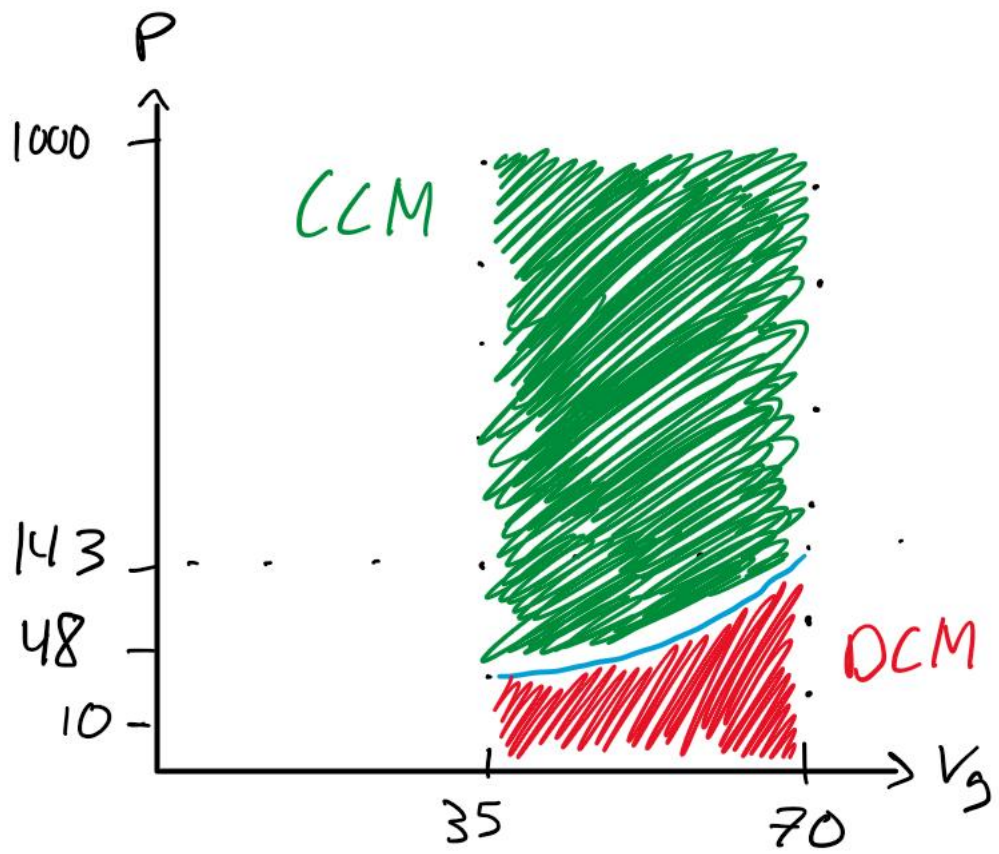
$$\boxed{P_{boundary} = \frac{V^2 T_s}{2L} \cdot \left(1 - \frac{V}{V_s}\right)}$$

$$P_{boundary}(35V) = \frac{28^2}{2.22 \cdot 10^{-6} \cdot 75 \cdot 10^3} \cdot \left(1 - \frac{28}{35}\right)$$

$$P_{boundary}(V_s) = 237.576 \cdot \left(1 - \frac{28}{V_s}\right)$$

$$P_{boundary}(35V) = 47.5W = 16.5\Omega$$

$$P_{boundary}(70V) = 142.5W = 5.5\Omega$$


P
1000
143
48
10
CCM
DCM
35
70
Vg

b) In DCM, D is lower for the same conversion ratio.

So use DCM conversion ratio to find D_{min}

Use CCM conversion ratio to find D_{max}

$$M_{CCM} = D_{CCM} = \frac{V}{V_g} \quad D_{CCM_{max}} = \frac{28V}{35V} = 0.8$$

$$M_{DCM} = \frac{V}{V_g} = \frac{2}{1+\sqrt{1+\frac{4K}{D^2}}}$$

Smallest K should create smallest D, but we'll check if voltage matters.

$$\text{smallest } K_{min} = \frac{2L}{R_{max} T_s} = 0.04$$

$$\frac{28}{V_g} = \frac{2}{1+\sqrt{1+\frac{4(0.04)}{D^2}}}$$

Try $V_{g\,min} = 35V$

$$0.8 = \frac{2}{1+\sqrt{1+\frac{0.16}{D^2}}}$$

$$0.8 + 0.8\sqrt{1+\frac{0.16}{D^2}} = 2$$

$$1.5 = \sqrt{1+\frac{0.16}{D^2}}$$

$$2.25 = 1 + \frac{0.16}{D^2}$$

$$1.25 = \frac{0.16}{D^2}$$

$$D^2 = \frac{0.16}{1.25}$$

$$D_{35V} = \sqrt{\frac{0.16}{1.25}}$$

$$D_{min}(35V) = 0.3578$$

Try $V_{g\,max} = 70V$

$$0.4 = \frac{2}{1+\sqrt{1+\frac{0.16}{D^2}}}$$

$$0.4 + 0.4\sqrt{1+\frac{0.16}{D^2}} = 2$$

$$4 = \sqrt{1+\frac{0.16}{D^2}}$$

$$16 = 1 + \frac{0.16}{D^2}$$

$$15 = \frac{0.16}{D^2}$$

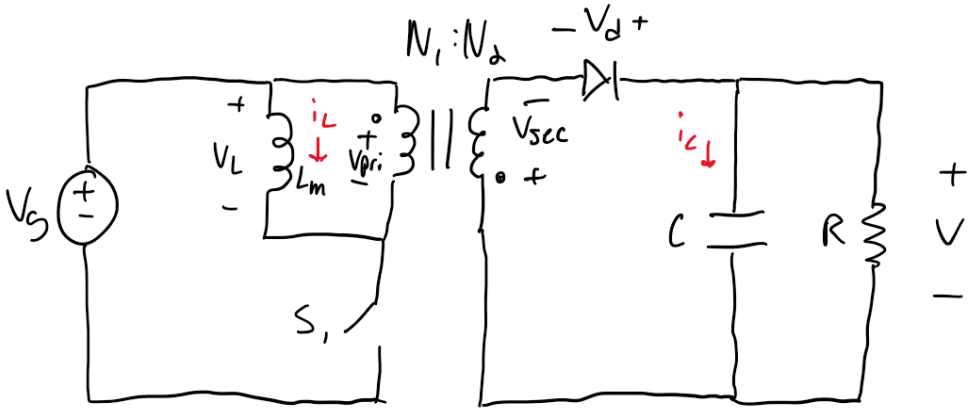
$$D_{70V} = \sqrt{\frac{0.16}{15}}$$

$$D_{min}(70V) = 0.1033$$

$$0.1033 \leq D \leq 0.8$$

70V @ 10W is deeper in DCM ∴ farther from the CCM duty ratio for the same V/V_g

Ericksom 6.2: Analysis of the DCM flyback converter



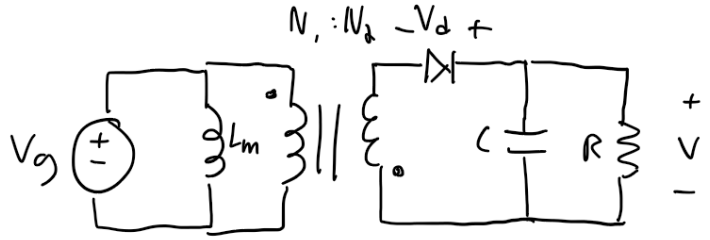
Tasks: a) sketch the converter circuits during the 3 states

b) Find the conditions needed to operate in DCM

c) derive the DCM conversion ratio (V/V_g) and the duration of the second state D_2

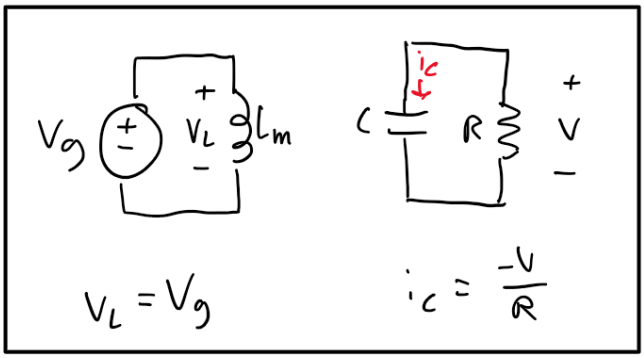
a)

State 1 : S_1 ON : $0 < t < DT_s$



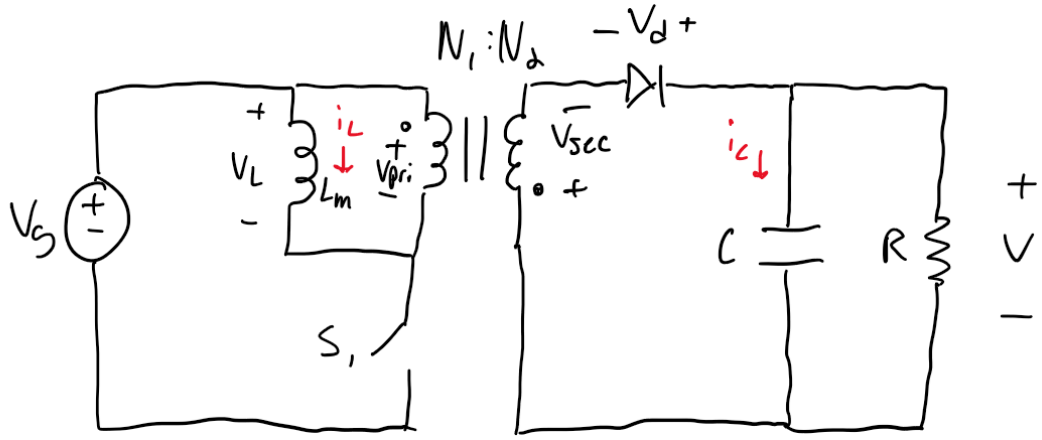
$V_L = V_{pri} = V_g \quad V_{sec} = \frac{N_2}{N_1} V_g \quad V_d = V + \frac{N_2}{N_1} V_g$

$\therefore i_d = 0$



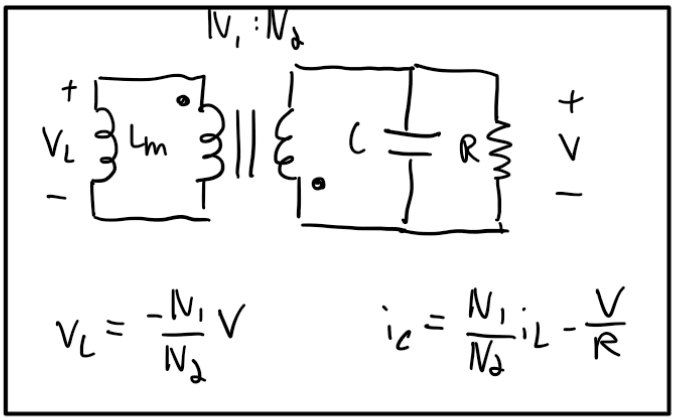
$V_L = V_g \qquad i_C = \frac{-V}{R}$

State 2 : S_1 OFF, D conducting : $0 T_s < t < (D+D_2) T_s$



$$V_L = -\frac{N_1}{N_2} V \qquad V_{sec} = -V \qquad V_d = 0$$

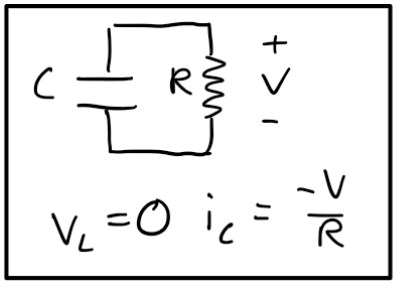
$$\therefore i_d = \frac{N_1}{N_2} i_L$$



$$V_L = -\frac{N_1}{N_2} V \qquad i_c = \frac{N_1}{N_2} i_L - \frac{V}{R}$$

State 3 : S_1 off, D not-conducting : $(D+D_2) T_s < t < T_s$

$i_L = 0$, $V_L = 0$



$$V_L = 0 \qquad i_c = \frac{-V}{R}$$

c) Find $M = \frac{V}{V_g}$ in DCM and what D_2 is equal to

technique:

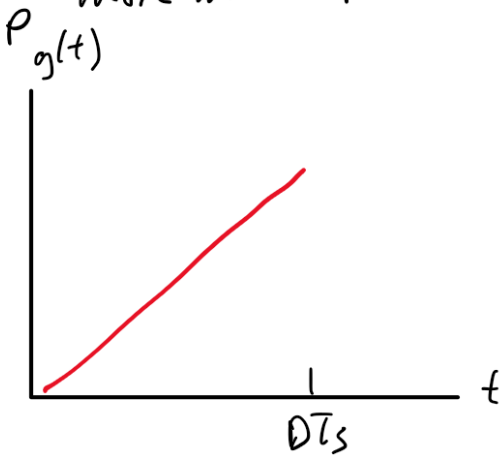
cool way uses power instead of volt-second balance and charge balance

Important to think outside of the box and use fundamental laws and intuition more than recipes.

Input power during state 1:

$$i_g(t) = i_L(t) = \frac{V_g}{L_m} \cdot t$$

$$P_g(t) = V_g \cdot i_g(t) = \frac{V_g^2}{L_m} \cdot t$$



$$E_{input} = \int_0^{DT_s} P_g(t)\,dt$$

$$E_{input} = \int_0^{DT_s} \frac{V_g^2}{L_m} \cdot t\,dt$$

$$= \frac{V_g^2}{L_m} \cdot \left[\frac{1}{2}(DT_s)^2 - 0\right]$$

$$E_{input} = \frac{D^2 T_s^2 V_g^2}{2L_m} \quad J/\text{period}$$

$$P_{input} = \frac{E_{input}}{T_s} \quad \frac{J/\text{period}}{\text{seconds/period}} = \frac{J}{\text{second}} = W$$

$$P_{input} = \frac{D^2 T_s V_g^2}{2L_m} \qquad P_{out} = \frac{V^2}{R}$$

$P_{out} = P_{input}$ assuming lossless conversion

$$\frac{V^2}{R} = \frac{D^2 T_s V_g^2}{2L_m}$$

$$\frac{V^2}{V_g^2} = D^2 \cdot \frac{RT_s}{2L_m}$$

DCM

$$\frac{V}{V_g} = D \cdot \sqrt{\frac{RT_s}{2L_m}}$$

CCM

$$\frac{V}{V_g} = \frac{N_2}{N_1} \cdot \frac{D}{1-D}$$

NOTE: $\frac{N_1}{N_2}$ does not contribute to DCM conversion ratio, but it does for CCM

To find the duration of state 2: D_2,

inductor volt-second balance:

$$\langle V_L \rangle = 0 = \frac{1}{T_s} \int_0^{T_s} V_L \, dt$$

$$0 = \frac{1}{T_s} \cdot \left[\int_0^{DT_s} V_g \, dt + \int_{DT_s}^{(D+D_2)T_s} -\frac{N_1}{N_2} V \, dt \right]$$

$$0 = \frac{1}{T_s} \cdot \left[DV_g T_s - D_2 \frac{N_1}{N_2} V T_s \right]$$

$$D_2 \frac{N_1}{N_2} V = D V_g$$

$$D_2 = \frac{N_2}{N_1} \cdot D \cdot \frac{V_g}{V}$$

$$D_2 = \frac{N_2}{N_1} \cdot \frac{D}{D\sqrt{\frac{RT_s}{2Lm}}}$$

$$\boxed{D_2 = \frac{N_2}{N_1} \cdot \sqrt{\frac{2Lm}{RT_s}}}$$

b) From notes, flyback CCM

conversion ratio is $\frac{V}{V_g} = \frac{N_2}{N_1} \cdot \frac{D}{1-D}$

at boundry, $M_{CCM} = M_{DCM}$

$$\therefore \quad \frac{N_2}{N_1} \cdot \frac{D}{1-D} = D \cdot \sqrt{\frac{RT_s}{2Lm}}$$

$$\frac{N_2}{N_1} \cdot \frac{1}{1-D} = \sqrt{\frac{RT_s}{2Lm}}$$

$$\frac{N_1}{N_2} \cdot (1-D) = \frac{1}{\sqrt{\frac{RT_s}{2Lm}}}$$

$$\frac{N_1}{N_2} - \frac{1}{\sqrt{\frac{RT_s}{2Lm}}} = \frac{N_1}{N_2} D_{BCM}$$

$$1 - \frac{N_2/N_1}{\sqrt{\frac{RT_s}{2Lm}}} = D_{BCM}$$

$$\boxed{DCM: \quad D < 1 - \frac{N_2/N_1}{\sqrt{\frac{RT_s}{2Lm}}}}$$

N_2/N_1 helps set the boundary